R3.1 Proton transfer reactions

Brønsted—Lowry acids and bases

Learning outcomes

By the end of this section you should be able to deduce the Brønsted—Lowry acid and base in a reaction.

Acidic substances give away a proton to other substances that can accept them. A hydrogen atom that has lost an electron is a proton.

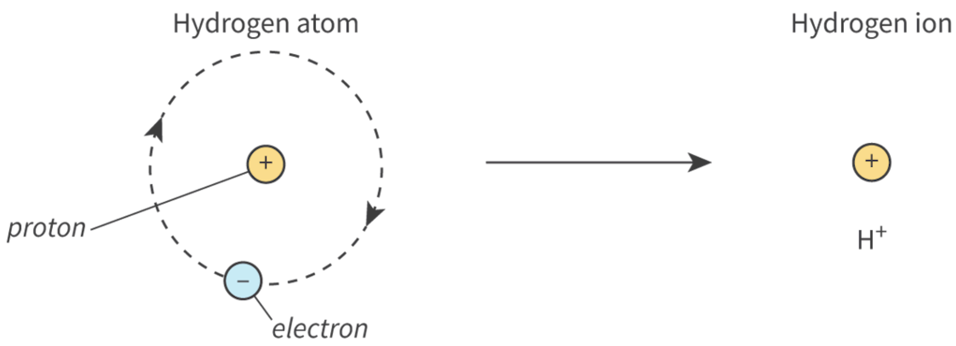

Figure 1. A hydrogen ion is a proton.

More information for figure 1

The diagram illustrates the transformation of a hydrogen atom into a hydrogen ion. On the left, it depicts a hydrogen atom consisting of a proton labeled with a positive sign in the center, encircled by a dashed line, representing an electron orbit with a negatively labeled electron. On the right side, an arrow leads to a simplified representation of a hydrogen ion, shown as just a proton with a positive charge symbol (H+). The diagram indicates the loss of the electron, resulting in the formation of a hydrogen ion.

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The Brønsted–Lowry theory defines acids and bases as either proton donors or proton acceptors. According to the theory:

- a Brønsted–Lowry acid donates a proton (is a proton donor)
- a Brønsted–Lowry base accepts a proton (is a proton acceptor)

A proton refers to a hydrogen ion H^+ which exists as the hydronium ion H_3O^+ in aqueous solution.

$$H^+(aq) + H_2O(l) \rightarrow H_3O^+(aq)$$

Study skills

A hydrogen ion in aqueous solution can be represented as $H^+(aq)$ or $H_3O^+(aq)$.

Nature of Science

Aspect: Theories

Why has the definition of acid evolved over time?

The central observable properties of acids and bases have been known for many hundreds of years. Acid solutions have a sour taste and react with a range of substances such as metal oxides and carbonates. Alkali solutions have a soapy feel and a bitter taste.

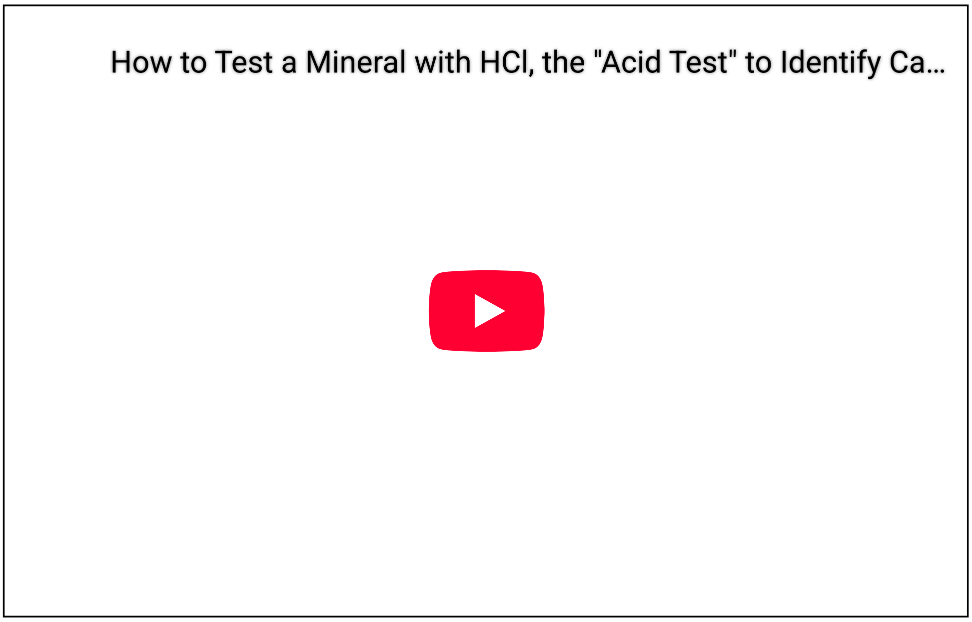
Lavoisier was the first to attempt to give a chemical rationale to the common properties of acids by suggesting that they all contained oxygen. It is true that many common acids are indeed oxyacids, but this idea was negated by the identification of hydrocyanic acid, HCN, and hydrochloric acid, HCl.

The German chemist, Justus von Liebig, first put forward the idea that all acids contained hydrogen in 1830, and also that this hydrogen could be replaced by metals in reactions.

Subsequent studies showed that acids, alkalis and salts could be grouped together as electrolytes — their solutions all conducted electricity and contained ions. The observation of a range of acids and alkalis, together with the products from their reactions, led to an explanation of their nature in terms of the ions present in their solutions.

Spectroscopic evidence has demonstrated that it is the hydronium ion, H_3O^+ (a protonated water molecule), which is the active ion present in acidic solutions (Brønsted-Lowry theory). These studies give substance to the ionic theory of acid—base behaviour.

A lot of reactions in life involve the transfer of protons from substances called acids to substances called bases. Did you know, for example, that one of these reactions is used by geologists to prove that a mineral is made of calcite? Watch **Video 1** to see how.



Video 1. The test for calcite.

The bubbling is the direct result of a proton, or a hydrogen cation, being transferred from the acid to the calcium carbonate producing carbon dioxide gas, water and calcium chloride. Can you relate this reaction to the one happening when acid rain falls on material that is made of limestone or calcium carbonate as seen in subtopic S3.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43116/)? Indeed, this same transfer of protons also occurs in the clouds, when water reacts with acidic fumes from the combustion of fossil fuels as seen in subtopic R1.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43475/). Then this is also what happens when the resulting acidic rain reacts with marble statues, or when an increasingly acidic ocean reacts with coral reefs, or when you pour vinegar on baking powder to produce ‘volcanoes’.

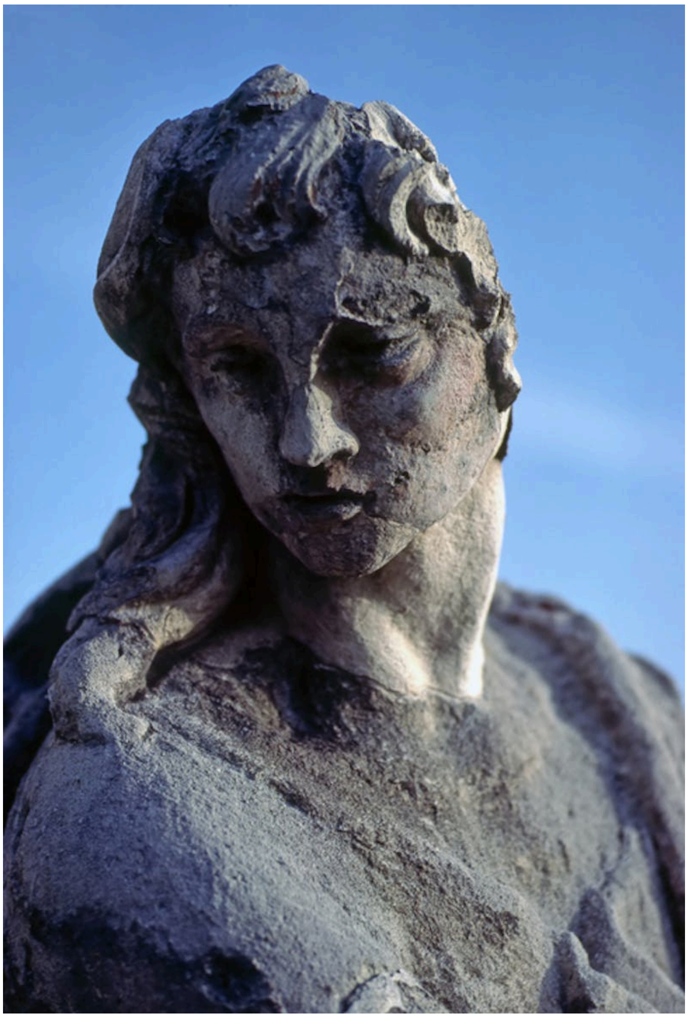


Figure 2. The effect of acid rain on monuments.

Credit: julof90, Getty Images

When hydrogen chloride gas dissolves in water, it donates a proton (H^+) to the water molecule, which accepts the proton.

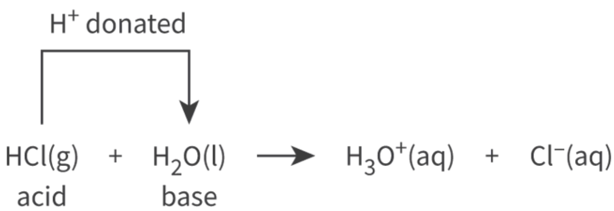


Figure 3. The reaction between hydrogen chloride gas and water.

More information for figure 3

The image depicts a chemical diagram illustrating the reaction between hydrogen chloride gas (HCl) and water (H_2O) to form hydronium ions (H_3O^+) and chloride ions (Cl^-). The diagram includes labeled components. HCl(g), labeled as 'acid', and H_2O(l), labeled as 'base', are the reactants shown on the left. An arrow labeled 'H^+ donated' points from above HCl(g) to H_2O(l), indicating the donation of a proton. To the right of the arrow indicating the direction of the reaction, the products are shown as H_3O^+(aq) and Cl⁻(aq). This visual representation illustrates the Brønsted—Lowry acid-base reaction concept, where HCl acts as an acid by donating a proton to water, forming a hydronium ion.

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The hydrogen chloride gas is acting as a Brønsted–Lowry acid and the water molecule as a Brønsted–Lowry base. The solution formed is named hydrochloric acid and is an example of a Brønsted–Lowry acid.

Study skills

All Brønsted—Lowry acids contain hydrogen, as they must be able to donate a hydrogen ion to another species.

Substances that can accept a proton are called Brønsted–Lowry bases. Ammonia (NH_3) is an example of a Brønsted–Lowry base. When ammonia reacts with water, it accepts a proton from the water molecule which acts as a Brønsted–Lowry acid.

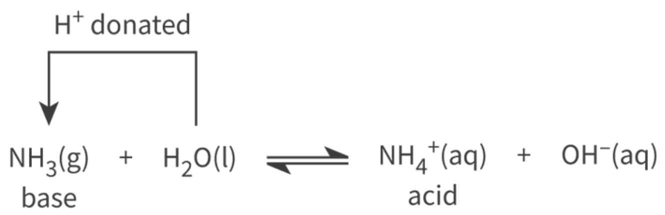

Figure 4. The reaction of ammonia and water.

More information for figure 4

The image is a diagram illustrating the chemical reaction between ammonia (NH_3) and water (H_2O). It shows ammonia acting as a Brønsted—Lowry base by accepting a proton (H^+) from water. The left side of the equation shows NH_3 in gaseous form and H_2O in liquid form. An arrow labeled "H^+ donated" indicates the transfer of a proton from H_2O to NH_3. This results in the formation of NH_4^+ (ammonium ion) and OH^- (hydroxide ion) on the right side of the equation. NH_4^+ is labeled as an acid, showing its potential to donate a proton in subsequent reactions, while OH^- remains as the conjugate base. The chemical equilibrium is represented by a double arrow between the reactants and products.

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Study skills

All Brønsted—Lowry bases must have a lone pair of electrons to form a bond with the hydrogen ion donated to them by the Brønsted—Lowry acid.

Names and formulae of common acids

In this DP chemistry course you will learn about three types of acid:

- binary acids
- oxyacids
- organic acids

Binary acids (see section S2.1.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/)) are formed when a halide reacts with hydrogen.

Table 1. Binary acids

Name of acid (IUPAC name)	Formula
Hydrofluoric acid (aqueous hydrogen fluoride)	HF(aq)
Hydrochloric acid (aqueous hydrogen chloride)	HCl(aq)
Hydrobromic acid (aqueous hydrogen bromide)	HBr(aq)
Hydroiodic acid (aqueous hydrogen iodide)	HI(aq)

Oxyacids (see subtopic S2.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43108/)) are derived from polyatomic ions.

Table 2. Oxyacids acids.

Name of acid	Formula
Nitric acid (aqueous hydrogen nitrate)	$HNO_3(aq)$
Sulfuric acid (aqueous hydrogen sulfate)	$H_2SO_4(aq)$
Carbonic acid (aqueous hydrogen carbonate)	$H_2CO_3(aq)$
Phosphoric acid (aqueous hydrogen phosphate)	$H_3PO_4(aq)$

Organic acids (see subtopic S3.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43122/)) contain a carboxyl group.

Table 3. Organic acids.

Name of acid	Formula
Ethanoic acid	$CH_3COOH(aq)$
Benzoic acid	$C_6H_5COOH(s)$

Although acids donate a hydrogen ion, not all substances that contain hydrogen behave as acids. For example, methane (CH_4) contains four hydrogen atoms, but does not donate any of them. Therefore, methane does not behave as an acid.

Acids can be classified as monoprotic, diprotic or triprotic depending on the number of hydrogen ions or protons that they can donate in an acid-base reaction.

Hydrochloric acid and nitric acid are monoprotic acids because they can only donate one hydrogen ion in acid-base reactions. Note that ethanoic acid is also monoprotic despite having four hydrogen atoms as only one of these is donated when ethanoic acid reacts with a base such as ammonia, NH_3.

Sulfuric acid is an example of a diprotic acid because it can donate two hydrogen ions in acid-base reactions.

Phosphoric acid is an example of a triprotic acid because it can donate three hydrogen ions in acid-base reactions.

Names and formulae of common bases

Many different substances are thought of as being bases. Some of the most common are discussed below.

Metal hydroxides and oxides (see section R3.1.7 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/reactions-of-acids-id-46056/)) that combine with water are bases. The names and formulae of some common bases are shown in **Table 4**.

Table 4. Oxides and hydroxide bases.

Name of base	Formula
Calcium oxide	$CaO(s)$
Copper(II) oxide	$CuO(s)$
Sodium hydroxide	$NaOH(aq)$
Lithium hydroxide	$LiOH(aq)$
Calcium hydroxide	$Ca(OH)_2(aq)$
Barium hydroxide	$Ba(OH)_2(aq)$

Carbonates and metal hydrogencarbonates are also examples of weak bases (see section R3.1.7 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/reactions-of-acids-id-46056/)). Some examples are shown in **Table 5**.

Table 5. Carbonates and hydrogen carbonates.

Name of base	Formula
Sodium hydrogen carbonate	$NaHCO_3(aq)$
Sodium carbonate	$Na_2CO_3(aq)$

Ammonia, amines and amides are also examples of weak bases (see section R3.1.7 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/reactions-of-acids-id-46056/)).

Table 6. Ammonia, amines and amides.

Name of base	Formula
Ammonia	$NH_3(aq)$
Ethanamine	$CH_3CH_2NH_2(aq)$

Study skills

Many bases are insoluble — they do not dissolve in water. However, if a base does dissolve in water, it is known as an alkali.

Examples of alkalis include the group 1 metal hydroxides sodium hydroxide and lithium hydroxide. Examples of insoluble bases include the metal oxides calcium oxide and copper(II) oxide.

Worked example 1

Write the equation for the reaction of:

1. Ethanamine with water
2. Nitric acid with water

1. Ethanamine is a base, so it accepts a proton from water:

$$CH_3CH_2NH_2 + H_2O \rightarrow CH_3CH_2NH_3^+ + OH^-$$

2. Nitric acid is an acid, it donates a proton to water:

$$HNO_3 + H_2O \rightarrow NO_3^- + H_3O^+$$

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:**
  - Thinking skills — Being curious about the natural world; Applying key ideas and facts in new contexts
  - Social skills — Actively seeking and considering the perspective of others

- **Time required to complete activity:** 20 minutes
- **Activity type:** Individual activity

Instructions:

1. Go on a hunt around your house and find at least five acids and two bases.
2. In a spreadsheet, list them in the first column of a table with three columns. State their common name and where they are found.
3. In the second column, add their scientific names.
4. In the third column, add their formulas.